

5.5 Substitution Worksheet

1. Use a u -substitution to compute, if possible, each of the following.

a) $\int_0^2 \frac{7x}{\sqrt{3x^2+5}} dx$

First we note that the function is continuous on $[0, 2]$, so we can apply FTCP2.

Let $u = 3x^2 + 5$ and thus $du = 6x dx \rightarrow \frac{1}{6} du = x dx$

Our limits of integration become: $u(0) = 3(0)^2 + 5 = 5$ & $u(2) = 3(2)^2 + 5 = 17$.

$$\int_0^2 \frac{7x}{\sqrt{3x^2+5}} dx = \int_5^{17} \frac{1}{6} \cdot \frac{7}{\sqrt{u}} du = \frac{7}{6} \int_5^{17} u^{-1/2} du = \frac{7}{6} \cdot 2u^{1/2} \Big|_5^{17} = \left[\frac{7}{3} \sqrt{17} - \frac{7}{3} \sqrt{5} \right]$$

b) $\int 3e^{3x} dx$

Let $u = 3x$ and thus $du = 3dx$.

$$\int 3e^{3x} dx = \int e^u du = e^u + C = e^{3x} + C$$

c) $\int_0^{\pi/3} \tan^3(x) \sec^2(x) dx$

First we note that the function is continuous on $\left[0, \frac{\pi}{3}\right]$, so we can apply FTCP2.

Let $u = \tan(x)$ and thus $du = \sec^2(x) dx$

Our limits of integration become: $u(0) = \tan(0) = 0$ & $u\left(\frac{\pi}{3}\right) = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$.

$$\int_0^{\pi/3} \tan^3(x) \sec^2(x) dx = \int_0^{\sqrt{3}} u^3 du = \frac{u^4}{4} \Big|_0^{\sqrt{3}} = \frac{(\sqrt{3})^4}{4} - \frac{(0)^4}{4} = \frac{9}{4}$$

d) $\int 2x \sin(x^2) dx$

Let $u = x^2$ and thus $du = 2x dx$.

$$\int 2x \sin(x^2) dx = \int \sin(u) du = -\cos(u) + C = -\cos(x^2) + C$$

e) $\int_0^{10} \frac{\ln(\sqrt{x})}{\sqrt{x}} dx$

We note that $f(x) = \frac{\ln(\sqrt{x})}{\sqrt{x}}$ is NOT continuous on $[0, 10]$, specifically because it is undefined at $x = 0$.

Therefore, the FTC2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

f) $\int_0^{\pi/2} 2 \cos(x) e^{\sin(x)} dx$

First we note that the function is continuous on $\left[0, \frac{\pi}{2}\right]$, so we can apply FTC2.

Let $u = \sin(x)$ and thus $du = \cos(x) dx$.

Our limits of integration become: $u(0) = \sin(0) = 0$ & $u\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$.

$$\int_0^{\pi/2} 2 \cos(x) e^{\sin(x)} dx = \int_0^1 2e^u du = 2e^u \Big|_0^1 = 2e^1 - 2e^0 = 2e - 2$$

g) $\int x^2 \sqrt{2+x^3} dx$

Let $u = 2 + x^3$ and thus $du = 3x^2 dx \rightarrow \frac{1}{3} du = x^2 dx$.

$$\int x^2 \sqrt{2+x^3} dx = \int \frac{1}{3} \sqrt{u} du = \frac{1}{3} \int u^{1/2} du = \frac{1}{3} \left(\frac{2}{3} u^{3/2} + C \right) = \frac{2}{9} u^{3/2} + C = \frac{2}{9} (2+x^3)^{3/2} + C$$

h) $\int_e^{e^4} \frac{dx}{x \sqrt{\ln(x)}}$

First we note that the function is continuous on $[e, e^4]$, so we can apply FTC2.

Let $u = \ln(x)$ and thus $du = \frac{1}{x} dx$.

Our limits of integration become: $u(e) = \ln(e) = 1$ & $u(e^4) = \ln(e^4) = 4$.

$$\int_e^{e^4} \frac{dx}{x \sqrt{\ln(x)}} = \int_1^4 \frac{1}{\sqrt{u}} du = 2u^{1/2} \Big|_1^4 = 2\sqrt{4} - 2\sqrt{1} = 4 - 2 = 2$$

i) $\int 3x^3 \sqrt{x^2+2} dx$

Let $u = x^2 + 2$ and thus $du = 2x dx \rightarrow \frac{1}{2} du = x dx$.

Also $u - 2 = x^2$.

$$\begin{aligned} \int 3x^3 \sqrt{x^2+2} dx &= \int 3(u-2) \sqrt{u} \frac{1}{2} du = \frac{3}{2} \int (u-2) u^{1/2} du = \frac{3}{2} \int (u^{3/2} - 2u^{1/2}) du = \frac{3}{2} \left(\frac{2}{5} u^{5/2} - 2 \cdot \frac{2}{3} u^{3/2} + C \right) \\ &= \frac{3}{5} u^{5/2} - 2u^{3/2} + C = \frac{3}{5} (x^2+2)^{5/2} - 2(x^2+2)^{3/2} + C \end{aligned}$$

j) $\int \frac{\ln(x)}{x} dx$

Let $u = \ln(x)$ and thus $du = \frac{1}{x} dx$.

$$\int \frac{\ln(x)}{x} dx = \int u du = \frac{1}{2} u^2 + C = \frac{1}{2} [\ln(x)]^2 + C$$

k) $\int_{-\pi/4}^{\pi/4} [x^3 + x^4 \tan(x)] dx$

First we note that the function is continuous on $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$, so we can apply FTCP2.

Notice that our limits of integration are opposite values, so we can check to see whether $f(x) = x^3 + x^4 \tan(x)$ is an even or odd function.

$$f(-x) = (-x)^3 + (-x)^4 \tan(-x) = -x^3 - x^4 \tan(x) = -[x^3 + x^4 \tan(x)] = -f(x)$$

So we see that $f(x)$ is an odd function.

Therefore, $\int_{-\pi/4}^{\pi/4} [x^3 + x^4 \tan(x)] dx = 0$.

2. Suppose f is continuous everywhere and $\int_1^4 f(x) dx = 5$. What is the value of $\int_0^{\sqrt{3}} 4x \cdot f(x^2 + 1) dx$?

First we note that the function is continuous on $[0, \sqrt{3}]$, so we can apply FTCP2.

Let $u = x^2 + 1$ and thus $du = 2x dx$

Our limits of integration become: $u(0) = (0)^2 + 1 = 1$ & $u(\sqrt{3}) = (\sqrt{3})^2 + 1 = 4$.

$$\int_0^{\sqrt{3}} 4x \cdot f(x^2 + 1) dx = \int_1^4 2f(u) du = 2 \int_1^4 f(u) du = 2(5) = 10$$

3. Find the average value of the function $f(x) = \frac{e^{\sqrt{x}}}{\sqrt{x}}$ on the interval $[25, 81]$.

$$f_{avg} = \frac{1}{81-25} \int_{25}^{81} \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

Since $f(x) = \frac{e^{\sqrt{x}}}{\sqrt{x}}$ is continuous on $[25, 81]$ we can use the FTC2.

Let $u = \sqrt{x}$ and thus $du = \frac{1}{2\sqrt{x}} dx \rightarrow 2du = \frac{1}{\sqrt{x}} dx$.

Our limits of integration become: $u(25) = \sqrt{25} = 5$ & $u(81) = \sqrt{81} = 9$.

$$f_{avg} = \frac{1}{81-25} \int_{25}^{81} \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = \frac{1}{56} \int_5^9 2e^u du = \frac{1}{28} e^u \Big|_5^9 = \frac{1}{28} [e^9 - e^5]$$

4. Suppose that f is an odd function and g is an even function which are each integrable on the interval $[-1, 1]$. Given that $\int_0^1 f(x) dx = 4$ & $\int_0^1 g(x) dx = 5$, evaluate the following definite integrals.

a) $\int_{-1}^1 [2f(x) + 3g(x)] dx$

$$\begin{aligned} \int_{-1}^1 [2f(x) + 3g(x)] dx &= \int_{-1}^1 2f(x) dx + \int_{-1}^1 3g(x) dx = 2 \int_{-1}^1 f(x) dx + 3 \int_{-1}^1 g(x) dx \\ &= 2(0) + 3 \cdot 2 \int_0^1 g(x) dx = 0 + 6(5) = 30 \end{aligned}$$

b) $\int_{-1}^1 [7 + 3(f(x))^9] dx$

$$\int_{-1}^1 [7 + 3(f(x))^9] dx = \int_{-1}^1 7 dx + \int_{-1}^1 3(f(x))^9 dx = 7[1 - (-1)] + 3 \int_{-1}^1 (f(x))^9 dx = 14 + 3 \int_{-1}^1 (f(x))^9 dx$$

Since $f(x)$ is an odd function, $(f(x))^9$ is an odd function:

$$\text{Let } h(x) = (f(x))^9. \text{ Then, } h(-x) = (f(-x))^9 = (-f(x))^9 = -(f(x))^9 = -h(x).$$

Therefore, we have that $\int_{-1}^1 [7 + 3(f(x))^9] dx = 14 + 3 \int_{-1}^1 (f(x))^9 dx = 14 + 3(0) = 14$.

5. Determine the average value of the function $f(x) = xe^{x^2}$ on the interval $[0, 2]$.

$$f_{\text{avg}} = \frac{1}{2-0} \int_0^2 xe^{x^2} dx$$

Let $u = x^2$ and thus $du = 2x dx \rightarrow \frac{1}{2} du = x dx$.

Our limits of integration become: $u(0) = (0)^2 = 0$ & $u(2) = (2)^2 = 4$.

$$f_{\text{avg}} = \frac{1}{2-0} \int_0^2 xe^{x^2} dx = \frac{1}{2} \int_0^4 \frac{1}{2} e^u du = \frac{1}{4} e^u \Big|_0^4 = \frac{1}{4} e^4 - \frac{1}{4} e^0 = \frac{1}{4} e^4 - \frac{1}{4} = \frac{e^4 - 1}{4}$$

6. Use a u -substitution to compute, if possible, each of the following.

a) Evaluate $\int_0^{1/2} \frac{6}{4x^2 + 1} dx$.

$$\int_0^{1/2} \frac{6}{4x^2 + 1} dx = \int_0^{1/2} \frac{6}{(2x)^2 + 1} dx$$

Let $u = 2x$ and thus $du = 2dx$.

Our limits of integration become: $u(0) = 2(0) = 0$ & $u\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) = 1$.

$$\int_0^{1/2} \frac{6}{(2x)^2 + 1} dx = \int_0^1 \frac{3}{(u)^2 + 1} du = 3 \tan^{-1}(x) \Big|_0^1 = 3 \tan^{-1}(1) - 3 \tan^{-1}(0) = 3\left(\frac{\pi}{4}\right) - 3(0) = \frac{3\pi}{4}$$

b) $\int x^3 \tan^2(x^4) dx$.

Let $u = x^4$ and thus $du = 4x^3 dx \rightarrow \frac{1}{4} du = x^3 dx$.

$$\int x^3 \tan^2(x^4) dx = \frac{1}{4} \int \tan^2(u) du = \frac{1}{4} \int [\sec^2(u) - 1] du = \frac{1}{4} [\tan(u) - u] + C = \frac{1}{4} [\tan(x^4) - x^4] + C$$

c) $\int_0^1 x e^{-x^2} dx$

First we note that the function is continuous on $[0,1]$, so we can apply FTCP2.

Let $u = -x^2$, then $du = -2x dx \Rightarrow -\frac{1}{2} du = x dx$.

Our limit of integration become: $u(0) = -(0)^2 = 0$ and $u(1) = -(1)^2 = -1$

Therefore, we have:

$$\int_0^1 x e^{-x^2} dx = \int_0^{-1} e^u \cdot \frac{-1}{2} du = -\frac{1}{2} \int_0^{-1} e^u du = -\frac{1}{2} e^u \Big|_0^{-1} = -\frac{1}{2} e^{-1} + \frac{1}{2} e^0 = \frac{1}{2} \left[1 - \frac{1}{e} \right]$$

d) $\int_{\frac{1}{2}}^2 \frac{\sec^{-1}(x)}{|x|\sqrt{x^2-1}} dx$

We note that $f(x) = \frac{\sec^{-1}(x)}{|x|\sqrt{x^2-1}}$ is NOT continuous on $\left[\frac{1}{2}, 2\right]$, specifically because it is undefined on the

interval $\left[\frac{1}{2}, 1\right]$ since on this interval $x^2 - 1 \leq 0$. Therefore, the FTCP2 does not apply and we have no way to determine such an area. We cannot proceed.

e) $\int \sqrt{2x+1} dx$

Let $u = 2x+1$ and thus $du = 2dx \rightarrow \frac{1}{2} du = dx$.

$$\int \sqrt{2x+1} dx = \int \frac{1}{2} \sqrt{u} du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{3} (2x+1)^{3/2} + C$$

f) $\int_1^2 t^2 (2t-1)^6 dt$

First we note that the function is continuous on $[1, 2]$, so we can apply FTC2.

Let $u = 2t - 1$, then $du = 2dt \Rightarrow \frac{1}{2} du = dt$ and we note that $u = 2t - 1 \Rightarrow t^2 = \frac{1}{4}(u+1)^2$.

Our limit of integration become: $u(1) = 2(1) - 1 = 1$ and $u(2) = 2(2) - 1 = 3$

Therefore, we have:

$$\begin{aligned} \int_1^2 t^2 (2t-1)^6 dt &= \int_1^3 \frac{1}{4}(u+1)^2 u^6 \cdot \frac{1}{2} du = \frac{1}{8} \int_1^3 (u^2 + 2u + 1) u^6 du = \frac{1}{8} \int_1^3 (u^8 + 2u^7 + u^6) du \\ &= \frac{1}{8} \left[\frac{1}{9} u^9 + 2 \cdot \frac{1}{8} u^8 + \frac{1}{7} u^7 \right] \Big|_1^3 = \frac{1}{8} \left[\frac{1}{9} (3)^9 + \frac{1}{4} (3)^8 + \frac{1}{7} (3)^7 \right] - \frac{1}{8} \left[\frac{1}{9} (1)^9 + \frac{1}{4} (1)^8 + \frac{1}{7} (1) \right] \\ &= \frac{1}{8} \left[\frac{3^7}{1} + \frac{3^8}{4} + \frac{3^7}{7} \right] - \frac{1}{8} \left[\frac{127}{252} \right] = \frac{1}{8} \left[\frac{283^7}{1} + \frac{73^8}{4} + \frac{43^7}{7} \right] - \frac{1}{8} \left[\frac{127}{252} \right] \\ &= \frac{1}{8} \left[\frac{115911}{28} \right] - \frac{1}{8} \left[\frac{127}{252} \right] \\ &= \frac{32596}{63} \end{aligned}$$

g) $\int_{-2}^0 \frac{42x}{(x^2+3)^2} dx$

First we note that the function is continuous on $[-2, 0]$, so we can apply FTC2.

Let $u = x^2 + 3$ and thus $du = 2x dx$.

Our limits of integration become: $u(-2) = (-2)^2 + 3 = 7$ & $u(0) = (0)^2 + 3 = 3$.

$$\int_{-2}^0 \frac{42x}{(x^2+3)^2} dx = \int_7^3 \frac{21}{u^2} du = \int_7^3 21u^{-2} du = -21u^{-1} \Big|_7^3 = -\frac{21}{u} \Big|_7^3 = -\frac{21}{3} - \left(-\frac{21}{7} \right) = -7 + 3 = -4$$

h) $\int_{-\pi}^{\pi} x \sqrt{x^2 + \cos(x)} dx$

First we note that the function is continuous on $[-\pi, \pi]$, so we can apply FTC2.

Notice that our limits of integration are opposite values, so we can check to see whether

$f(x) = x \sqrt{x^2 + \cos(x)}$ is an even or odd function.

$$f(-x) = (-x) \sqrt{(-x)^2 + \cos(-x)} = -x \sqrt{x^2 + \cos(x)} = -f(x)$$

So we see that $f(x)$ is an odd function and therefore $\int\limits_{-\pi}^{\pi} x\sqrt{x^2 + \cos(x)}dx = 0$